

Image Processing based Image to Cartoon Generation: Reducing complexity of large computation arising from Deep Learning

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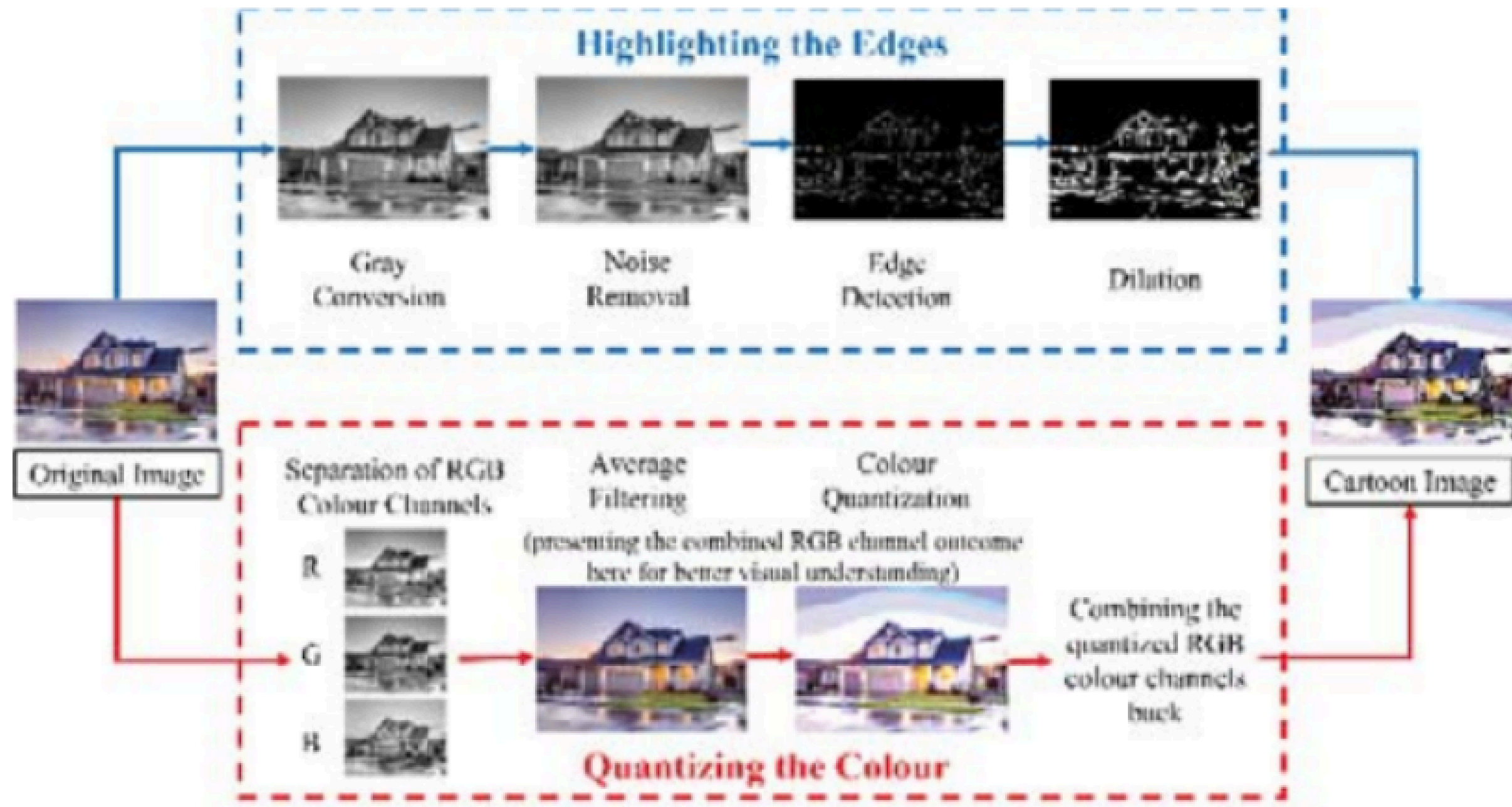
INTRODUCTION

- Cartoon avatars are a popular way for self-expression in the digital world.
- The growing demand for personalized cartoon images an efficient algorithm is needed to automate the process and quickly convert 1000's of images without human interventions
- Image processing techniques offer a simpler and fast alternatives to ML methods and deep learning methods like GAN for cartoon image generation ,avoiding the complexities, time consumption and training data requirements of the models
- This paper proposes an efficient algorithm using image processing techniques for fast and automated cartoon image generation.

Objective

- Develop a computationally efficient algorithm for cartoon image generation using image processing techniques, eliminating the need for deep learning.
- In this paper aims to develop an algorithm in MATLAB to convert an image into cartoon ,preserving the visual quality of cartoon images with sharp edges, smooth color regions, and contextual details of the original image.

Methodology



Noise Removal

Applied a median filter to remove noise while preserving edges, avoiding the averaging filter due to its blurring effect.

Edge Detection

Used canny edge detector for precise edge extraction. Also check with sobel, prewitt, robert and log edge detector

Morphological Dilation

Enhanced edges for a bold cartoon effect using morphological dilation with structuring elements at 0° (focused on horizontal edges) and 90° (focused on vertical edges).

Image Smoothing

Split the original image into Red Green Blue channels. Applied averaging filter on RGB channels to smooth and reduce pixel variations.

Color Quantization

Reduced color complexity by mapping pixel values to a specific range using the formula:

$P = [p/(a-b)] * (a+b/2)$. Divided 256 pixel levels into a 6 smaller set of mapped values

Combining Channels

Combined quantized images of all three RGB channels to reconstruct a 3D matrix .

Edge Integration

Merged dilated edges with the quantized image, assigning black pixel values (0) to edge regions for edge enhancement.

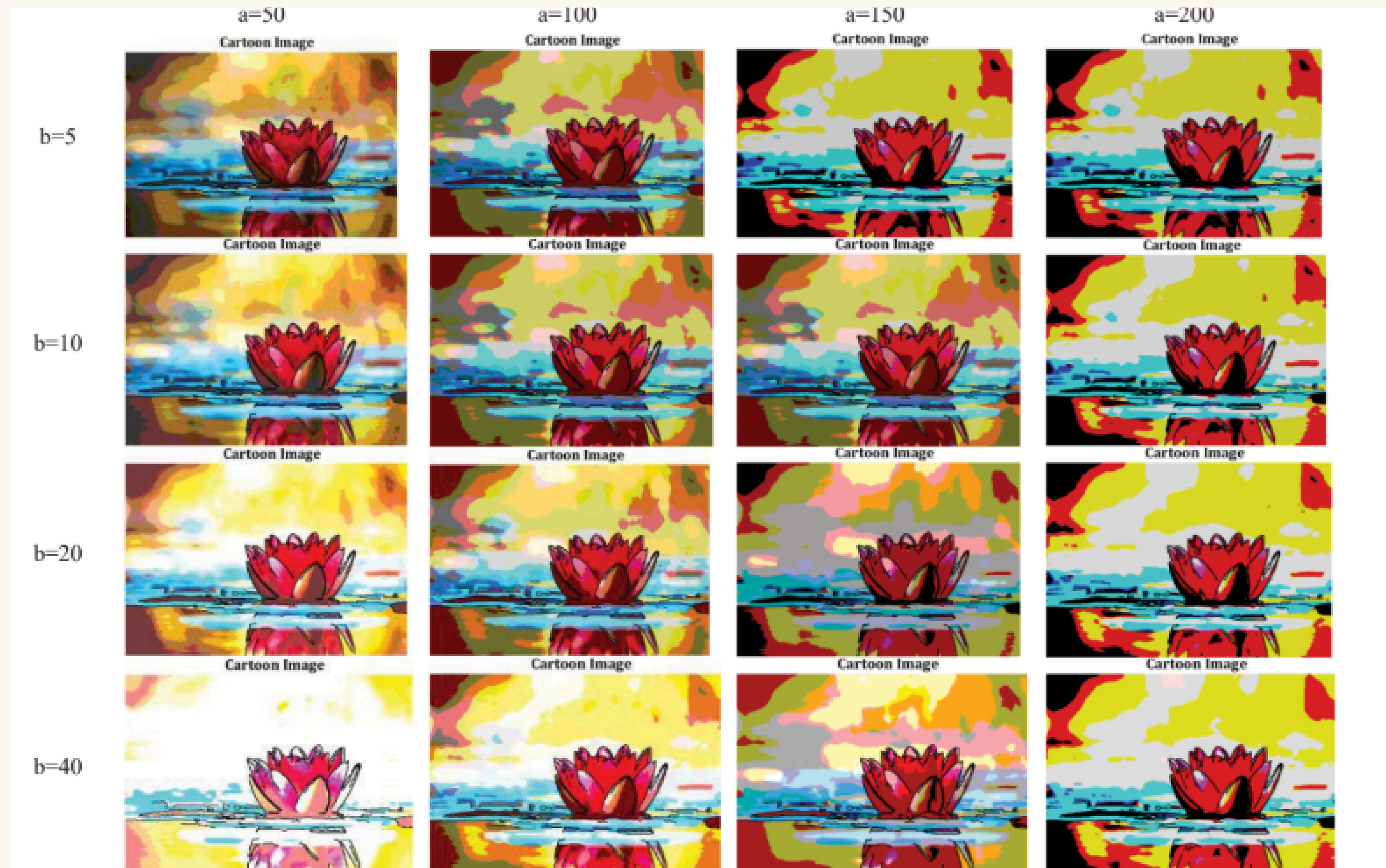
Final Cartoon Image Generation

Produced a cartoon image with bold edges and reduced color palette, creating a visually appealing output.

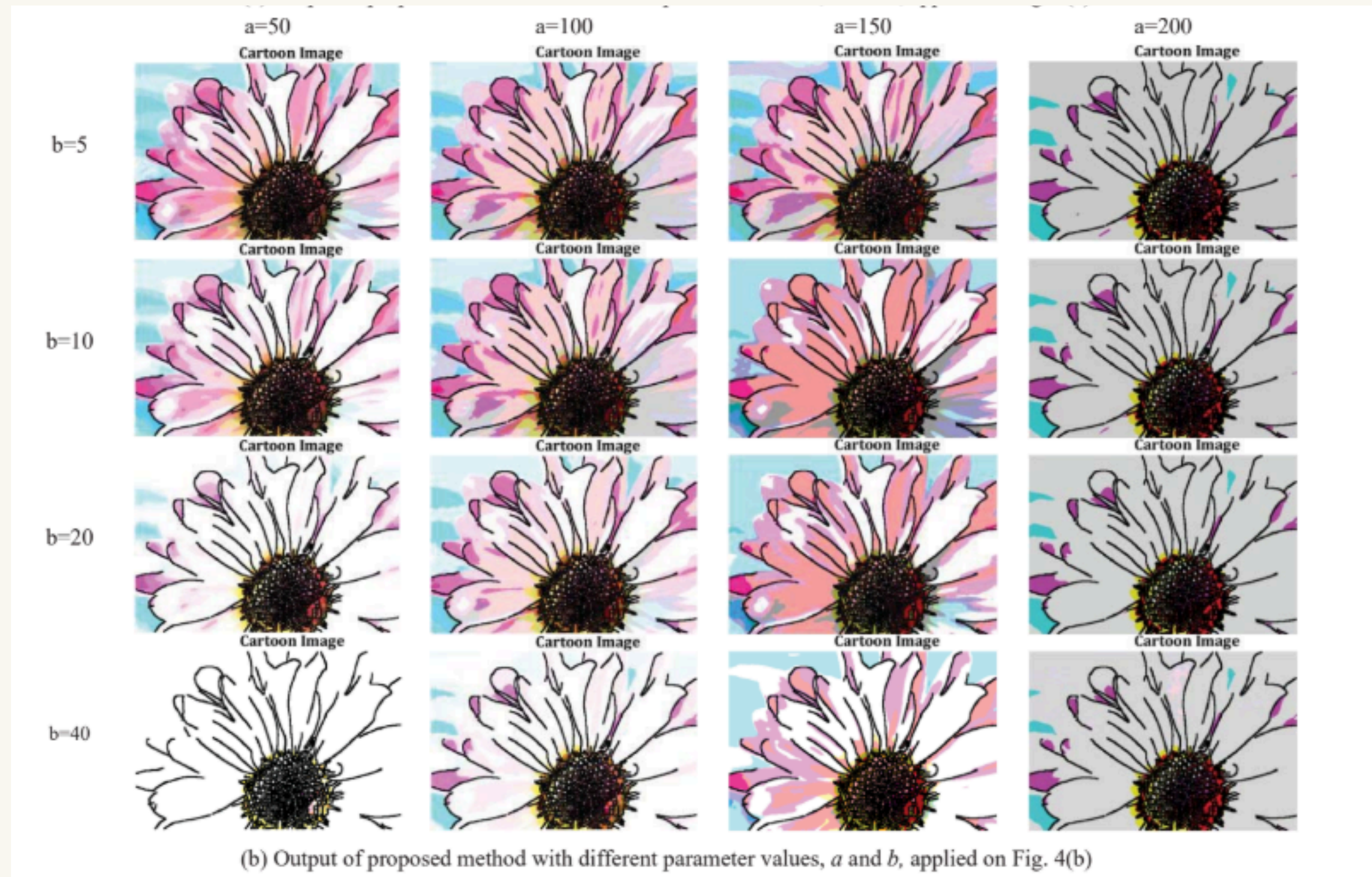
Experimental Analysis

- The proposed method turns images into cartoons using canny edge detection, average filtering, and color quantization, with a and b adjusting the colors and edges.
- a (50–200) and b (5, 10, 20, 40) were varied to study their effects.
- Larger b values resulted in overexposed images as pixel value shifted to higher intensity ranges.
- Larger a values led to underexposed images as pixel value shifted to lower intensity ranges..
- We get the optimal parameter $a=50$ and $b=10$ cartoon images achieves balanced effect with bold edges, clear shapes, and 6–8 color levels, creating the ideal cartoon style.
- The proposed method created sharper, higher-contrast cartoon images with clearer edges, outperforming traditional and deep learning methods.

Experimental Analysis



Experimental Analysis



Conclusion

- The proposed method generates cartoon images using color quantization, edge detection, and dilation to enhance image quality.
- It produces high-quality cartoon images with better contrast and color quantization compared to existing image processing methods, while requiring less computation and time.
- The approach can also be applied to video sequences for generating cartoon videos.

Reference

- [1] S. K. Shrivastava and R. Gajjar, "Image Processing Based Image to Cartoon Generation: Reducing Complexity of Large Computation Arising from Deep Learning," Electronics and Communication Engineering Department, Institute of Technology, Nirma University, Ahmedabad, Gujarat, India, 2024.
2023 International Conference on Computational Intelligence and Sustainable Engineering Solutions (CISES)

THANKYOU

